Martinez-Blasco *et al. Financial Innovation* (2025) 11:126
https://doi.org/10.1186/s40854-025-00867-9

Financial Innovation

RESEARCH Open Access

Mapping innovation tools and financial inclusion: a bibliometric analysis

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Abstract

This study employs bibliometric and visualisation techniques to analyse global trends in financial inclusion and the innovative tools that promote it. By examining a multi-database compilation of 4202 documents sourced from Scopus and Web of Science, the significant role of book chapters in disseminating research in this field is highlighted. Our analysis reveals rapid growth in publications, particularly from China and India, and identifies key influential works and authors. Additionally, it is observed that financial inclusion is approached from a multidisciplinary perspective. The findings of this study also indicate a shift in research focus from traditional concepts such as "housing" and "banking" to "digitalisation" and "sustainability." Key research trends in financial innovative tools include regulatory frameworks, mobile money, and financial education. The issues discussed in this article seek to contribute to the ongoing dialogue on developing intellectual frameworks withing the financial inclusion literature. They offer valuable insights for policymakers, industry practitioners, and researchers in the fields of financial innovation and inclusion.

Keywords: Financial inclusion, Bibliometric analysis, WoS, Scopus, Co-citation, Co-occurrence

JEL Classification: G21, G23, G28

Introduction

This paper contributes to the growing literature on financial inclusion (FI) by conducting a comprehensive literature review of the topic. It explores trends in innovative tools that support the expansion of FI. According to Sarma's (2008) definition, FI entails ensuring that every individual in a country can readily and conveniently participate in the formal financial system, including access to and use of financial services. The World Bank (2014), for its part, defines FI as the ratio of households and companies using financial services. FI is considered to be a key driver of economic growth, poverty reduction, and social stability. Empirical evidence indicates that it significantly contributes to economic development and poverty reduction, particularly in developing economies (Beck et al. 2007a, b, Demirgüç-Kunt et al. 2018). Its impact also extends to transition (Bayar et al. 2021; Banerjee et al. 2023) and developed economies (Demirgüç-Kunt et al. 2017; Manta et al. 2023).

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Several factors, such as high financial costs, inadequate distribution networks, inappropriate risk analysis mechanisms, and regulatory systems that are not well suited (Demirgüç-Kunt 2008; Schuetz and Venkatesh 2020; Saifurrahman and Kassim 2024), hinder FI among low-income segments of the population. This may be due to an insufficient supply of financial services rather than to a lack of demand (Prior and Argandoña 2009; Geraldes et al. 2022). As a result, there is an unmet demand (Abraham 2018) for financial services in this low-income population. Low levels of FI may be caused by the factors listed below.

First, the lack of financial depth in developing countries is closely tied to the high fees banks charge (Demirgüç-Kunt et al. 2017). Operating efficiency ratios show a correlation between higher financial margins and lower operating efficiency (Beck et al. 2007a, b). This inefficiency is widespread across developing regions, with sub-Saharan Africa, Eastern Europe, and Latin America exhibiting higher financial margins (Matousek et al. 2015). The lack of competition within national financial systems contributes to the high cost of financial services (Aterido et al. 2011).

Second, the low density of bank branches in developing countries results in limited access to banking services for the population (Dabla-Norris et al. 2015). This is primarily due to the inability of most potential clients to generate sufficient income to support the fees of traditional banks. Managing bank branches in developing countries is more expensive than in developed countries because of security concerns and deficient infrastructure (Batista et al. 2020). Moreover, inequalities exist in developing countries, and regions with greater purchasing power enjoy better access to bank branches (Sarma and Pais 2011).

Third, inappropriate credit risk analysis methodologies also explain the low levels of FI in developing nations (Beck and Brown 2019). Credit risk analysis processes in developing countries rely heavily on assessing the expected cash flows of borrowers. However, these processes primarily consider formal income sources, which poses a challenge for a significant portion of the population that relies on the informal sector (Beck and Cull 2014). Another aspect of risk analysis involves linkage variables, which measure the client's relationship with the financial institution. In countries with limited access to banking services, these variables only apply to a small percentage of the population, further hindering access to credit. Greater emphasis on sociodemographic data could expand access while maintaining controlled delinquency ratios (Karim and Tarazi 2016).

Finally, the regulatory landscape of the financial sector in developing countries is fragmented and challenging to navigate. This has led numerous institutions to operate outside formal legal frameworks (Dabla-Norris et al. 2015). Strengthening and formalizing this sector could have significant positive effects on economic development, which is tainted by corruption (Spyromitros and Panagiotidis 2022). Demirgüç-Kunt and Levine (2001) and Levine (2005) support this view. De Koker (2011) and Zeqiraj et al. (2022) identify key questions that should be addressed to provide guidance regarding the alignment of anti-money laundering, combatting the financing of terrorism, promoting institutional quality, and FI objectives. Moreover, high regulatory costs give rise to negative efficiency ratios in traditional commercial banks operating in developing countries. It is hence expensive to provide basic financial services to low-income segments. Insufficient

regulation (Allen et al. 2016) also makes it difficult for microfinance entities to operate under proper oversight.

Because of the abovementioned factors, unbanked customers are compelled to rely on more expensive and less efficient informal services, hampering their economic progress (Beck et al. 2000). Facilitating FI requires overcoming persistent barriers to financial access through innovative financial tools that reduce the costs of financial services, expand distribution networks, and adopt risk methodologies suitable for developing countries. Key solutions include remittance flows, new business models, decentralized technologies, regulatory improvements, and financial education. These measures align the provision of financial services with societal demand (Cruz et al. 2023), thus facilitating widespread FI.

Building on this framework, Prior (2010) identifies eight financial innovation tools (FITs) that may address the challenges related to accessing financial services: mobile money, nonbank financial distribution, behavioral credit risk, workers' remittances, adapted financial regulation, financial education, microfinance business models, and blockchain technology. Mobile money products are the most cost-effective and efficient means of meeting the financial needs of unbanked individuals (Munyegera and Matsumoto 2016; Abor et al. 2018; Li and Xu 2021). The use of agents enhances the density of the financial network (Tooker and Maurer 2016). The implementation of alternative credit risk methodologies enables individuals working in the informal economy to access credit and other fundamental financial services (Urrea and Maldonado 2011; Aitken 2017). Workers' remittances play a crucial role as primary sources of capital inflow in emerging economies and increase the efficiency with which financial institutions serve unbanked individuals (Anzoategui et al. 2014). Regulatory changes adapt the regulatory framework to the unique needs of the unbanked (Allen et al. 2016). Policies that focus on enhancing financial education enable consumers to access and use financial services while fostering increased trust in the financial system (Sarma and Pais 2011). Financial business models such as microfinance are tailored to better suit the financial needs of the economically disadvantaged (Mushtaq and Bruneau 2019). Integrating new technologies such as blockchain makes financial business models more streamlined and efficient (Arner et al. 2020; Chen and Bellavitis 2020).

Whereas FITs have improved access to financial services, a structured analysis of their role in FI research is lacking. Although prior bibliometric studies on FI have provided valuable insights, they are limited in both scope and depth. Existing analyses (Chhatoi et al. 2021; Gálvez-Sánchez et al. 2021; Li and Xu 2022; Priyan et al. 2023; Ghosh 2024; Suryavanshi et al. 2024) have focused on specific domains such as microfinance or financial literacy and have relied exclusively on a single database or excluded nonjournal sources, thereby omitting a substantial portion of the scientific output. They hence offer limited insight into the mechanisms through which innovation drives FI, and the role of specific FITs is largely overlooked. Recent studies (Echchakoui 2020; Kumpulainen and Seppänen 2022; Sahid et al. 2023; Suryavanshi et al. 2024) highlight the value of using combined databases.

To address these gaps, a bibliometric analysis of the scientific output of FI, with a particular focus on research trends in FITs that support its expansion, is performed in this study. Drawing on data from the two most widely used academic databases, Scopus and

Web of Science (WoS), the most comprehensive dataset on FI to date has been constructed. Two complementary approaches are used: performance analysis and science mapping (Noyons et al. 1999). The analysis systematically maps eight key FITs: mobile money, nonbank financial distribution, behavioral credit risk, workers' remittances, adapted financial regulation, financial education, microfinance business models, and blockchain technology. While these tools have been studied individually in the context of FI, their collective influence on FI research remains largely unexplored. This study seeks to bridge that gap by integrating the conceptual and empirical perspectives of FI research. As Ghosh (2024) argues, greater attention needs to be given to developing the intellectual structures of FI research. This paper contributes to that goal by offering a unified framework that identifies key publication trends, influential contributors, and underexplored thematic connections.

Because of the gaps identified in previous bibliometric studies, the first contribution of this research is theoretical. It extends the existing FI literature by integrating FITs into a unified analytical framework. Rather than considering these tools as separate areas of inquiry, this study examines their combined influence on the development of FI research. This integrated perspective provides a clearer understanding of how innovation shapes the conceptual foundations of FI, revealing thematic connections and underexplored intersections that can guide future theoretical advancements in the field.

The second contribution is empirical and involves the construction of a unique merged database for analysis, combining data from WoS and Scopus to serve a threefold purpose. First, this merged database may yield better results and enhance research quality. This point is supported by Kumar and Kumar (2023), Priyan et al. (2023), and Suryavanshi et al. (2024). Second, while previous bibliometric analyses of FI often rely on small sample sizes, as observed in Chhatoi et al. (2021), Gálvez-Sánchez et al. (2021), Li and Xu (2022), Priyan et al. (2023), Ghosh (2024), and Suryavanshi et al. (2024), this study employs the largest sample to date to analyze recent bibliometric trends in the field. Finally, by incorporating not only research articles but also other reliable sources, this study provides additional insights compared with previous research (Suryavanshi et al. 2024). Notably, disparities in results are observed when relying solely on WoS or Scopus rather than on a merged database combining both.

This study has practical implications for policymakers and practitioners and shows that a deeper understanding of FITs can inform the design of more effective policies, regulatory frameworks, and business strategies that promote both FI and sustainable development.

On the basis of the research gaps identified and on the theoretical and empirical contributions of this study, a set of research questions (RQs) was developed to guide the analysis. These questions explore the evolution of FI research, the intellectual structure of the field, the role of FITs, and potential directions for future studies. The formulation and sequence of the RQs follow the logic of the empirical analysis in this paper. The analysis begins with an overview of publication trends, followed by mapping the intellectual structure, an in-depth examination of FIT-related research, and finally, the identification of research gaps and emerging themes.

To understand the evolution of research in FI, this paper explores publication trends and identifies the most influential documents, sources, and authors in the field. RQ1

provides an overview of the field on the basis of a newly constructed merged database previously unexplored in the literature. Therefore, RQ1 is formulated as follows:

RQ1: What are the current publication trends, the most influential documents, sources, and authors in FI?

Given the importance of mapping the intellectual framework of a field, this study examines the intellectual structure of the FI literature to understand the evolution of the body of knowledge and the connections between the key themes. This leads to RQ2:

RQ2: What is the intellectual structure of the literature on FI?

Considering the importance of FITs in advancing FI, it is crucial to explore the research trends surrounding these tools to understand their role in facilitating FI. Thus, RQ3 is defined as:

RQ3: What are the research trends in FITs that promote FI?

Finally, this paper also seeks to identify potential areas for further research, particularly those that remain underexplored. Accordingly, RQ4 is as follows:

RQ4: What are the potential areas for future research?

The remainder of the paper is organized as follows: Section "Theoretical foundations and literature review" provides a brief overview of the definitions associated with the eight innovative tools that promote FI, along with a review of the bibliometric literature associated with each tool. Section "Methodology and sample description" describes the methodology employed and provides a brief description of the sample. Section "Results" presents the results of the performance analysis, scientific mapping, and evaluation of FITs. Finally, Section "Discussion and conclusions" discusses the most significant findings and addresses the research questions.

Theoretical foundations and literature review

The growing interest in FI has generated a diverse and rapidly expanding body of literature. The present section outlines both the theoretical and empirical foundations of the study. The section is divided into three parts. The first part introduces theoretical frameworks that explain the selection and analysis of FITs. The second part reviews bibliometric studies on FI, identifying trends, gaps, and methodological limitations. The third part focuses on the eight FITs, presenting their definitions and a contextualized review of how each tool functions across varying institutional, regulatory, and socioeconomic environments. It also summarizes the specific research associated with each tool.

Theoretical foundations

This study draws on three complementary theoretical frameworks to examine the relationship between FITs and FI: financial development theory, technology adoption models, and the inclusive growth framework. Together, they provide a multidimensional basis for understanding how FITs perform across different institutional and socioeconomic contexts. Financial development theory emphasizes the role of financial systems in promoting economic growth, investment efficiency, and poverty reduction. Within

this framework, FI is viewed as a component of financial deepening that enables broader participation in formal markets. A growing body of evidence links improved access to credit, savings, and insurance with higher investment and welfare gains, particularly in low-income settings (Beck et al. 2007a, b; Durusu-Ciftci et al. 2017). In this context, FITs such as agent banking, remittances, and microfinance serve as intermediating mechanisms through which financial development goals are achieved.

At the micro level, technology adoption theories offer insights into how individuals and institutions interact with digital financial services. The Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) highlight perceived usefulness, ease of use, and facilitating conditions as key drivers of user adoption. These models are especially relevant for understanding the uptake of tools such as mobile money, digital credit scoring, and blockchain-based services, which require users to adopt new digital behaviors (Venkatesh et al. 2003; Munyegera and Matsumoto 2016; Schuetz and Venkatesh 2020). The Diffusion of Innovations Theory sheds light on how innovations spread across social systems shaped by communication channels, adopter characteristics, and institutional readiness (Zanello et al. 2016).

However, another perspective, the inclusive growth framework, positions FI as a tool for achieving equitable development. Access to financial services is viewed not only as a means of increasing efficiency but also as a way to reduce inequality and promote social mobility (Zins and Weill 2016). In this view, the value of FITs lies in their ability to reach underserved populations, such as women, youth, and rural communities, and to support broader development goals when embedded in effective policy and institutional ecosystems. This aligns with recent empirical findings showing that inclusive finance contributes to poverty alleviation and inclusive growth, especially when supported by complementary structural reforms (Demirgüç-Kunt et al. 2018).

Review of bibliometric studies on FI

Bibliometric analysis has become an increasingly popular tool for understanding the development and structure of FI research. Several studies have contributed to this expanding body of work. Goyal and Kumar (2021) conducted an earlier effort to analyze publication patterns in FI, noting the rise of digital financial services and the policy push toward inclusive finance. Chhatoi et al. (2021) examined 463 Scopus-indexed articles and reported that research on financial services and microfinance in India gained strong momentum after 2013. Gálvez-Sánchez et al. (2021) offered thematic mapping of FI research, identifying digital money, fintech, and sustainability as emerging trends. Li and Xu (2022), working with the WoS database, revealed the growing interdisciplinarity of the field, particularly its integration in technology, development economics, and financial regulation.

More recent studies have further developed empirical mapping methods for the field. Priyan et al. (2023) explored emerging keywords, citation patterns, and collaborative author networks within FI. Ghosh (2024) conducted a coword analysis of 206 articles to stress the conceptual diffusion of FI research and its stakeholder relevance. Suryavanshi et al. (2024) presented one of the most comprehensive bibliometric assessments to date, analyzing over 2100 Scopus-indexed documents to uncover thematic clusters on digitalization, financial literacy, and geographic disparities.

Kumar and Kumar (2023) added depth through cocitation and keyword co-occurrence analyses, offering insight into the thematic fragmentation of the field. Similarly, Sang (2023) conducted a large-scale bibliometric study of 2,165 papers from 1998 to 2023, revealing key publishing trends, research trajectories, and international collaboration networks. Sangma and Das (2024) complemented this work by focusing on 356 Scopus articles published between 2013 and 2022, emphasizing the rise of digital finance, mobile money, and poverty-focused themes, particularly in the context of developing countries.

While these studies have provided valuable insights, several limitations remain. Many rely exclusively on a single bibliographic database, typically Scopus (Chhatoi et al. 2021; Gálvez-Sánchez et al. 2021; Priyan et al. 2023; Suryavanshi et al. 2024; Sang 2023; Sangma and Das 2024) or WoS (Li and Xu 2022), which narrows the scope of analysis and limits the inclusivity of their findings. Some studies explicitly exclude nonjournal sources, such as book chapters and conference proceedings—notably Gálvez-Sánchez et al. (2021), Suryavanshi et al. (2024), and Kumar and Kumar (2023)—despite the relevance of these sources in an interdisciplinary field such as FI. Although these studies provide rich descriptive insights, they often abstain from linking bibliometric patterns to conceptual frameworks or policy implications. This lack of integration between empirical mapping and theoretical insight is a recurring gap in the literature (Suryavanshi et al. 2024).

Beyond thematic trends and methodological choices, a major limitation in existing bibliometric studies on FI is their exclusive reliance on either Scopus or WoS, despite growing evidence that these databases yield different and, on occasion, contradictory results. Scopus tends to index a broader range of materials, including conference papers, book chapters, and publications from emerging economies, whereas WoS emphasizes high-impact journals, often in English and from Western institutions (Mongeon and Paul-Hus 2015). For example, Sangma and Das (2024), who resort to Scopus, identify the United States and the World Bank as dominant contributors, including research centered on poverty and economic growth. Li and Xu (2022), using WoS, draw attention to European institutions and regulatory themes. Citation counts and institutional rankings often differ depending on the database used, potentially distorting assessments of scholarly influence (Alryalat et al. 2019). Boshoff and Akanmu (2017) further show that WoS is biased toward international collaborations, whereas Scopus better captures regionally focused work in the Global South. These discrepancies have meaningful implications for a field such as FI, which is inherently multidisciplinary and globally distributed. Thus, the merged dataset in this study addresses these blind spots, offering a more balanced and inclusive empirical basis for understanding the evolution of FI research.

FITs: definitions

In this subsection, the eight FITs that aim to promote FI are defined. These tools have been carefully selected because of their potential to address various barriers to FI. The existing literature on bibliometric studies related to FITs, which have examined the scientific production of these tools, albeit often in isolation, has also been reviewed.

FIT1: mobile money

Mobile money services allow users to conduct a wide range of financial transactions-without the need for a traditional bank account (Maurer 2012). These transactions have the potential to significantly increase FI by providing individuals excluded from the formal banking system with access to financial services (Ozili 2018). Their success depends on several contextual factors, including mobile network coverage, digital trust, and regulatory flexibility. In East Africa, for instance, they were widely adopted owing to supportive financial regulations and high mobile penetration. However, parts of South China are lagging behind due to policy and infrastructure constraints (Munyegera and Matsumoto 2016; Abor et al. 2018; Li and Xu 2022).

FIT 2: nonbank financial distribution

Nonbank financial distribution networks refer to systems or infrastructures of financial service providers that are not traditional banks but play a critical role in delivering various financial products and services to consumers. These networks include microfinance institutions, mobile money providers, payment processors, fintech companies, cooperatives, and retail agents. They may significantly contribute to increasing FI by expanding access to financial services (Gabor and Brooks 2017). Countries such as Brazil and Kenya have benefited from clearly defined agent models, whereas weaker institutional oversight in other regions has limited their reliability and scale (Gabor and Brooks 2017; Demirgüç-Kunt et al. 2018).

FIT 3: Behavioral credit risk

Behavioral credit risk methodologies involve the use of behavioral and psychological factors to assess the creditworthiness of individuals or businesses. They go beyond traditional credit scoring models, which rely heavily on financial data, by considering non-traditional data sources and behavioral factors. These methodologies aim to provide a more holistic and inclusive approach to credit assessment. They are particularly useful in low-income contexts where formal credit history is lacking. However, regulatory frameworks for data privacy and consumer protection vary across countries (Óskardóttir et al. 2019; Chen et al. 2021).

FIT 4: Workers' remittances

When migrant workers send part of their earnings to their home country to support their families, these transfers are known as workers' remittances (Lee et al. 2021). They play a significant role in the global economy, as millions of people work abroad and send money home to their families (Misati et al. 2019). In sub-Saharan Africa and Southeast Asia, remittance-linked products have gained traction despite high transaction fees and regulatory frictions.

FIT 5: Adapted financial regulation

Regulatory measures and policies tailored to the specific needs and circumstances of underserved or marginalized populations with the goal of promoting FI are referred to as adapted financial regulation (Allen et al. 2016). These regulations recognize that traditional financial regulations may not be suitable for all segments of the population and

may represent a barrier to accessing financial services (Zins and Weill 2016). The scope and ambition of these regulations differ widely. For example, countries such as Rwanda and India have made rapid strides because of strong institutional leadership, whereas others face constraints from legacy regulations or a lack of political will.

FIT 6: Financial education

Financial education refers to the process of providing individuals with knowledge, skills, and resources to help them make informed financial decisions, manage their finances effectively, and achieve their financial goals. It encompasses a wide range of topics, including budgeting, saving, investing, debt management, understanding financial products, and financial planning (Goyal and Kumar 2021). Financial education can be provided through various channels, such as formal education programs, workshops, seminars, online courses, and informational materials. It helps individuals increase their trust in the financial system (Fungáčová et al. 2016).

FIT 7: Microfinance business models

Microfinance business models are financial service delivery approaches designed to provide small-scale financial services to individuals, households, and small businesses, often in underserved or economically disadvantaged communities (Ghosh and Bhandari 2013). These models aim to increase FI by offering a range of financial products and services to people who have limited access to traditional banking and financial institutions. For example, in South Asia, group-based models dominate, whereas in Latin America, commercial microfinance institutions have gained traction under more liberalized frameworks (Mushtaq and Bruneau 2019).

FIT 8: Blockchain technology

Blockchain technology is a decentralized and distributed digital ledger system that records transactions across multiple computers in a way that ensures transparency, security, and immutability of data. Transactions recorded on a blockchain are grouped into blocks and linked together in chronological order, forming a chain. Each block contains a list of transactions, and once added to the blockchain, it cannot be altered or deleted. This technology has the potential to increase FI in various ways (Schuetz and Venkatesh 2020). However, feasibility is shaped by digital infrastructure, regulatory clarity, and public–private coordination. While pilot programs in countries such as Estonia and Nigeria show promise, blockchain technology remains limited in jurisdictions where digital policy environments are lacking (Chen and Bellavitis 2020).

Literature review of bibliometric studies regarding FITs

Several bibliometric studies have assessed scientific production related to specific FITs and their impact on FI. The next paragraphs review recent bibliometric and scientometric research papers that focus on one or more FITs and their relationships with FI.

The study of Ali et al. (2023) is one of the recent works that analyses scientific production related to FI and microfinance business models. Their study highlights emerging research topics, such as financial technology, crowdfunding, and financial literacy. For FI and literacy, the studies of Goyal and Kumar (2021) and Kumar and Kumar (2023) show

that research on financial literacy has advanced considerably. Since 2020, Khongwir and Sharmiladevi (2023) have observed a shift in the research focus on FI and financial literacy. Traditional topics such as financial indicators and economic factors have given way to a stronger emphasis on digitalization. Rodríguez-López and Souto (2020) explore the literature on income-generating activities financed through microfinance institutions and argue that entrepreneurs need both financing and training to increase their capacity.

Liu et al. (2023) identify crowdfunding, peer-to-peer lending, and mobile banking as the most frequently discussed topics and suggest that blockchain and other emerging technologies hold promising potential for future research. Pal et al. (2021) also support this idea. This perspective is reinforced by Carè et al. (2024), who review mobile money, blockchain, and digital banking, identifying thematic clusters and emphasizing the contextual and technological dynamics influencing their adoption.

Several authors have noted an increase in scientific publications concerning FI and nonbank financial distribution. For instance, Ahmi et al. (2020) reported an increase in research output focusing on alternative technology firms that offer various financial services, such as mobile payments, and utilize innovative business models. Bhatt et al. (2022) noted that the rapid digitalization of financial services has increased research interest in this area. Furthermore, Afjal (2023), Liu et al. (2024), and Gera et al. (2024) identify financial technology and digital financing services among the most popular research areas. The findings of Molina-Collado et al. (2021) show that customer satisfaction, innovation, corporate social responsibility, the internet, and consumer acceptance were the main research themes from 2000 to 2020. Similarly, Li and Xu (2022) conclude that behavior analysis and household finance are the main research lines.

Finally, Radic et al. (2023) carried out the only bibliometric analysis on the relationship between remittances and economic growth. However, it did not address FI.

Methodology and sample description

Methodology

WoS and Scopus, the most generally accepted and comprehensive data sources for conducting bibliometric analysis, were used as research methodologies (Mongeon and Paul-Hus 2016). WoS, currently owned by Clarivate Analytics, is the leading and most extensively used database for research publications and citation analysis. It incorporates several citation indices, including the Science Citation Index Expanded (SCIE), covering reputable science and technology publications; the Social Science Citation Index (SSCI), a multidisciplinary index for social science journals; and the Emerging Sources Citation Index (ESCI). The latter includes journals in the fields of science, technology, and social sciences, with the potential to be incorporated into Journal Citation Reports (JCR) (Wang and Chen 2010; Gaviria-Marin et al. 2019). Scopus, launched by Elsevier in 2004, is also a multidisciplinary, prominent, and widely used bibliographic database in the academic and research communities. It has proven to be as reliable as WoS (Zhu and Liu 2020).

Searches were conducted in the WoS Core Collection and Scopus in November 2023 via the search term "Financial inclusion." The search parameters were "Article title, Abstract, and Keywords" in Scopus and "Topic" in WoS. No specific start date was set for the collection of documents, but an end date was fixed. Thus, the historical analysis

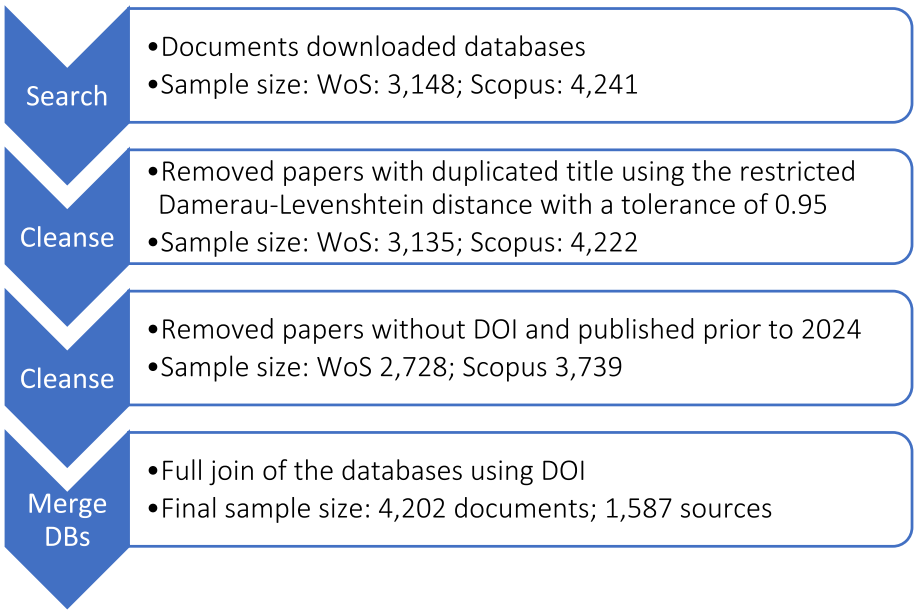

Fig. 1 Document identification and screening. *Source* Authors' own work

began when documents matching the search criteria were available in the databases. All available documents were downloaded in a CSV file.

The data files in this study were processed via a combination of two widely employed software tools:

R and Microsoft Excel (MS Excel). The use of R, a powerful open-source statistical software, facilitated advanced data manipulation, statistical analysis, and visualization. MS Excel was employed for its versatility in handling and organizing data. The bibliometric analysis was conducted via a combination of the *bibliometrix* R package (Aria and Cuccurullo 2017) and the authors' customizations. The analysis included a productivity assessment in which the evolution of publication numbers and their distribution across countries, sources employed to publish research, and authors were analyzed. The significance of publications and authors was determined by document productivity and citation counts. Word clouds generated from abstracts were used to identify changes in the publication landscape over time. To address the structure of the research topic, relational methods were employed. A bibliometric mapping approach was applied using VOSViewer software[1] (Van Eck and Waltman 2010), enabling the visualization of networks resulting from co-occurrence and cocitation analyses of documents. Conceptual structure mapping, through multiple correspondence analysis, was used to provide a visual representation of thematic clusters, allowing researchers to discern how different topics interrelate (Aria and Cuccurullo 2017).

For this study, all the outputs obtained from the databases were considered. Overlapping between the two databases was eliminated through the Digital Object Identifier (DOI). Figure 1 illustrates the process, starting from the downloaded documents until

[1] https://www.vosviewer.com/.

the final database of outputs is analyzed. The final sample comprises 4202 documents; 73.32% consist of research articles; 8.32% are book chapters; and 5.24% are conference papers.

In the final sample, 54.4% of the documents were from both Scopus and WoS, 34.8% were from Scopus, and the remaining 10.9% were from WoS. To standardize the information from the databases, the information retrieved from Scopus was used because of its higher document count. When information was missing from Scopus but available in WoS, the missing information was sourced from WoS. Regarding the citation count for each document, a similar approach was adopted. If a document appeared in both databases, the Scopus citation count was considered. For documents sourced solely from WoS without a Scopus citation count, a regression model was employed to predict the number of citations they would have in Scopus. This methodological approach aligns with previous research indicating a strong correlation between citation counts in both databases (Martín-Martín et al. 2018; Echchakoui 2020). The model used to predict the number of citations had an R^2 of 0.98. With respect to authors' citations, the number of citations in each database was analyzed separately. This approach mitigates potential distortions arising from database dominance, acknowledging that citation metrics can vary significantly owing to differences in how each database indexes and links citations (Ansorge 2023).

Sample description

Three dimensions were selected to describe the data analyzed: the distribution of scientific production per year, the productivity of the authors, and the countries concentrating the highest volume of published research.

Documents

The first documents on FI in the search were published in the late 1990s. Leyshon et al. (1998) and Fuller (1998) are considered pioneers in this topic. Other authors identified previous publications classified in the field of FI (Gálvez-Sánchez et al. 2021). However, owing to the lack of detailed parameters regarding the search criteria employed, it is impossible to conclusively confirm that their findings correspond to the usage of the term employed in this study.

Figure 2 shows an exponential increase in publications on FI from 1998 to 2023. While fewer than 60 documents appeared before 2010, output accelerated from 2011 onward, reaching over 1000 documents in 2023. The annual growth rate in the number of documents published during the entire period is 28.59%, whereas the average growth rate in scientific publications is estimated to be 4% per year (Bornmann et al. 2021). The global financial crisis and the use of ICTs in FI could be some of the reasons for this increase (Goyal and Kumar 2021; Gallego-Losada et al. 2023). According to Suryavanshi et al. (2024), the World Bank's commitment to ensuring equitable access to finance by 2030 also contributes to the increase in published articles. Similarly, Afjal (2023) claims that this growth can be attributed to swift progress in financial technology and the growing acknowledgment of FI's crucial role in fostering economic growth and reducing poverty.

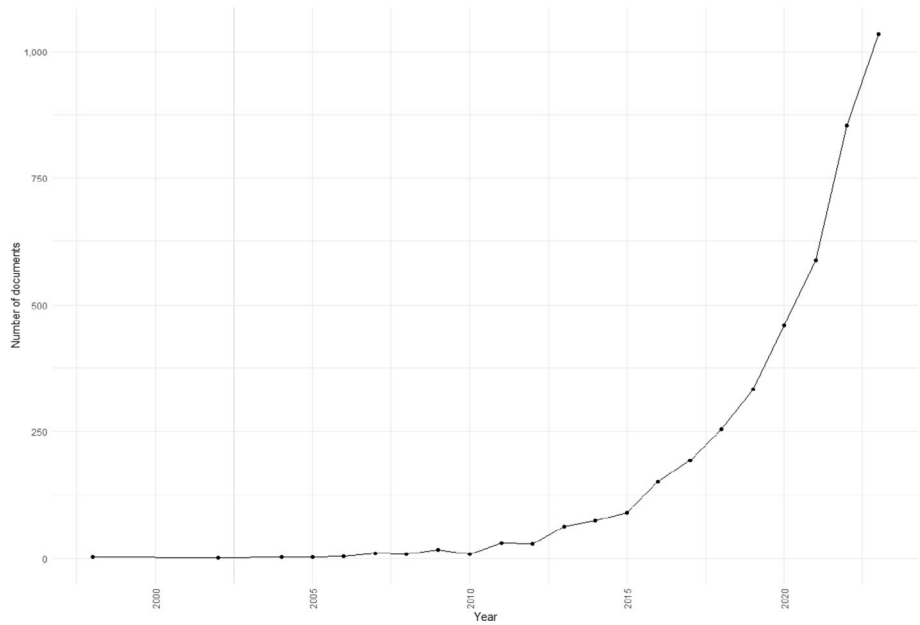

Fig. 2 Documents published per year. *Source* Authors' own work via WoS and Scopus data (2023)

Table 1 Number of documents published per author

Number of documents	Number of authors	Percentage (%)
1	6,107	80.30
2	896	11.78
3	276	3.63
4	135	1.78
5	63	0.83
6	50	0.66
7	22	0.29
8	15	0.20
9	7	0.09
10	8	0.11
11 to 34	26	0.34

Source: Authors' own work via WoS and Scopus data (2023)

Authors

In the literature reviewed in this paper, 7,605 authors made contributions. They collectively appear 10,708 times in the documents analyzed. A total of 770 documents were written by a single author. The dataset also stresses the collaborative nature of academic papers, as the average number of authors per document is 2.55 coauthors. This emphasizes the common practice of scholars working together on research projects. Importantly, 32.98% of the documents involve international collaboration. On average, each author contributed approximately 0.553 documents, which shows diverse research involvement across the scholarly community.

Following one of the main laws in bibliometrics established in Lotka (1926), Table 1 shows the authors' productivity patterns. The data emphasize that most authors

published only one study, accounting for 80.30% of all the documents. Only 7.93% of the authors contributed three or more documents. This means that there is a limited number of specialized authors, and a significant portion of authors are occasional contributors. The high number of occasional contributors reflects the broad appeal of FI research across disciplines. The small group of prolific authors suggests that specialized expertise is still developing. This pattern indicates an emerging field with growing but still fragmented scholarly engagement.

Countries

For geographical distribution, Fig. 3 shows that the countries that produce the most research on the topic are located in Asia. China and India take the lead in research production, contributing 413 and 295 documents, respectively. Developed nations such as the United States and the United Kingdom follow closely, with 288 and 200 documents, respectively. Other significant contributors to the global discourse on the subject include South Africa with 107 documents, Malaysia with 83, Australia with 80, Nigeria with 77, and Ghana with 68. It can be concluded that the most productive countries in which authors publish documents on FI are developing nations.

Results

This section is organized into two subsections. The first subsection presents findings on influential documents, sources, and authors, along with an analysis of the intellectual structure of the topic. In the second subsection, the analysis is directed specifically toward examining the tools that promote FI.

Influential documents, sources, and authors: intellectual structure

In this subsection, the results obtained from the most cited documents, together with details on the most cited sources and authors, are presented. This analysis provides an understanding of the researchers' contributions and the impact of their studies in the field. The field is also mapped through a cocitation analysis of documents. It graphically

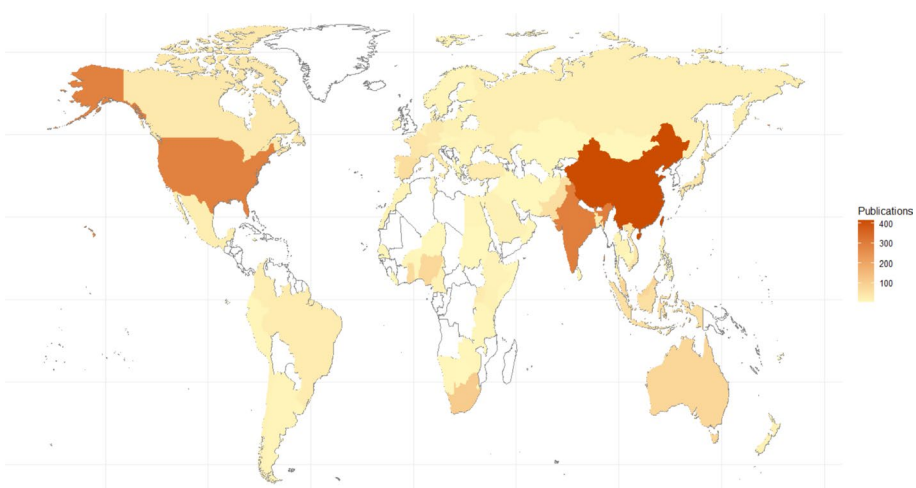

Fig. 3 Publications per country. *Source* Authors' own work via WoS and Scopus data (2023)

illustrates connections between authors, a word cloud analysis of the abstracts of the years selected, and a conceptual structure map.

Influential documents

Table 2 presents the ten most cited documents collected. The documents with more than 400 citations address digital finance, FI and development, and individual and country characteristics related to FI. Ozili (2018) examines issues related to optimizing the

Table 2 Most cited documents

Title	Author/s	Publication Year	Source	Times cited	Times cited per year
Impact of digital finance on financial inclusion and stability	Ozili, P.K	2018	Borsa Istanbul Review	547	91.2
Financial inclusion and development	Sarma, M.; Pais, J	2011	Journal of International Development	461	35.5
The foundations of financial inclusion: Understanding ownership and use of formal accounts	Allen, F.; Demirgüç-Kunt, A.; Klapper, L.; Martinez Peria, M.S	2016	Journal of Financial Intermediation	409	51.1
Predicting poverty and wealth from mobile phone metadata	Blumenstock, J.; Cadamuro, G.; On, R	2015	Science	390	43.4
Financial literacy: A systematic review and bibliometric analysis	Goyal, K; Kumar, S	2021	International Journal of Consumer Studies	329	109.7
The digital revolution in financial inclusion: international development in the fintech era	Gabor, D.; Brooks, S	2017	New Political Economy	328	46.9
Global Financial Development Report 2014-Financial inclusion	World Bank Group	2014	International Bank for Reconstruction and Development/ The World Bank Group	305	30.5
Does financial inclusion, renewable and non-renewable energy utilization accelerate ecological footprints and economic growth? Fresh evidence from 15 highest emitting countries	Usman, M.; Amjad Makhdum, M. S.; Kousar, R	2021	Sustain Cities Society	285	95
The determinants of financial inclusion in Africa	Zins, A.; Weill, L	2016	Review of Development Finance	271	33.9
Financial inclusion and economic growth in OIC countries	Kim, D-W.; Yu, J-S.; Hassan, M.K	2018	Research in International Business and Finance	271	45.2

Source: Authors' own work via WoS and Scopus data (2023)

effectiveness of digital finance for individuals, businesses, and governments, distinguishing financial data inclusion from FI, which contributes to the paper's popularity. Sarma and Pais (2011) study the relationship between FI and development, identifying country-specific factors associated with FI. Allen et al. (2016) analyze individual and country characteristics associated with FI and find that greater FI is related to lower banking fees, proximity to financial intermediaries, and stronger institutions. The most cited documents focus on digital finance, access to formal financial services, and development, suggesting that FI is increasingly viewed as a policy tool for economic growth. The high citation counts of these works show that they are foundational, and the emphasis on technology reflects a shift toward innovation-driven approaches.

Most frequent and cited sources

The 4202 documents analyzed were published in 1587 different sources. Table 3 presents an overview of the ten journals most frequently used to disseminate research findings regarding FI. These ten journals published only 10.47% of the total research production analyzed, which shows that publications in this area are widely dispersed. *Sustainability* emerges as the leading source, with 103 documents published on the topic. This reflects its significant role in academic discussions regarding FI. The *International Journal of Social Economics* is the second-highest contributor, followed closely by *Environmental Science and Pollution Research*. A closer look at the table shows the diverse academic disciplines that actively contribute to research production in FI. The disciplines most frequently observed in the research on FI include environmental sciences, economics, econometrics, business, and finance.

Metrics for sources exceeding 500 citations are presented in Table 4. The analysis carried out in this paper extends beyond absolute citation numbers and considers the number of documents published in the source. Using both metrics, a third column is included to show the position of each source among the 1,587 sources that constitute the sample, considering both the number of times cited and the number of documents published. *Sustainability* emerges in the top-ranking position. Other noteworthy journals include *World Development*, which is ranked 13th with a citation count of 1,405, and *Environmental Science and Pollution Research*, which is ranked 3rd, with 723 citations across 60 documents. The fact that the documents were spread across numerous journals confirms the interdisciplinary nature of FI research. The presence of high-ranking journals suggests that FI is gaining academic recognition, while the diversity of outlets reflects its broad policy and practical relevance. However, this dispersion also poses challenges for consolidating the field's knowledge base. The wide variety of publication outlets may lead to fragmentation, making it harder for researchers to identify core debates, methodologies, and leading contributors.

Most productive and most cited authors

Difficulties arise when assessing the authorship of documents, as a unique identifier is needed to properly assign documents to a single person. Identifiers such as names and surnames are not suitable for this task, as they can easily produce duplications. To address this issue and given the composition of the database, the Scopus Author ID was used. This identifier does not exist in WoS, which is why the results

Table 3 Top publication sources

Source	Number of documents	WoS Edition	Journal Impact Factor (JIF) Journal Citation Index (JCI) and Quartile—2022	Scimago Journal Rank Quartile-2022
Sustainability	103	SSCI	Environmental Studies-JIF Q2 Environmental Sciences-JIF Q2	Computer Networks and Communication-SJR Q2 Energy Engineering and Power Technology-SJR Q2 Environmental Science-SJR Q2 Geography, Planning and Development-SJR Q1 Hardware and Architecture-SJR Q2 Management, Monitoring, Policy and Law-SJR Q2 Renewable Energy, Sustainability and Environment-SJR-Q2
International Journal of Social Economics	62	ESCI	Economics-JCI Q2	Economics, Econometrics and Finance-SJR Q3 Social Science (miscellaneous)-SJR Q2
Environmental Science and Pollution Research	60	SCIE	Environmental Sciences-JIF Q1	Environmental Chemistry-SJR Q2 Health, Toxicology and Mutagenesis- SRJ Q1 Medicine (miscellaneous)-SRJ Q1 Pollution-SRJ Q1
Cogent Economics & Finance	43	ESCI	Economics-JCI Q2	Economics and Econometrics-SRJ Q3 Finance-SRJ Q3
Journal of Risk and Financial Management	36	ESCI	Business, Finance-JCI Q2	Accounting- SJR Q3 Business, Management and Accounting-SJR Q3 Economics and Econometrics-SJR Q3 Finance-SJR Q3
Lecture notes in Networks and Systems	32	–	–	Computer Networks and Communication-SJR Q4 Control and Systems Engineering-SJR Q4 Signal Processing- SJR Q4
Enterprise Development and Microfinance	29	–	–	Development-SJR Q4 Geography, Planning and Development-SRJ Q4
Applied Economics	28	SSCI	Economics-JIF Q2	Economics and Econometrics-SRJ Q2
Finance Research Letters	27	SSCI	Business, Finance-JIF Q1	Finance- SJR Q1
Journal of International Development	20	SSCI	Development Studies-JIF Q4	Development-SJR Q2 Geography, Planning and Development-SJR Q2

Source: Authors' own work via WoS and Scopus data (2023)

leave out the data from this database, excluding 461 documents. Although this decision might alter the results, it does not affect the overall ranking because the top 10 authors would remain unchanged if the data from WoS were included. Table 5 shows the most productive authors in the field and their respective affiliations, using only documents indexed in Scopus. The top contributors include Peterson K. Ozili from the University of Essex, England. George Okello Candiya Bongomin stands out for his 23 documents, and his colleagues from Makerere University in Uganda have also made significant contributions. Together, they form a highly productive team of researchers. In general, a significant number of researchers affiliated with institutions

Table 4 Most cited journals

Sources	Times cited	Number of documents	Times cited/Number of documents Position
World Development	1,405	23	13
Sustainability	1,245	103	1
Borsa Istanbul Review	907	19	19
Finance Research Letters	818	27	9
Environmental Science and Pollution Research	723	60	3
Journal of International Development	692	26	10
International Journal of Social Economics	587	62	2
Review of Development Finance	576	6	118
Emerging Markets Finance and Trade	562	17	26
New Political Economy	530	10	69
Research in International Business and Finance	529	14	42

Source: Authors' own work via WoS and Scopus data (2023)

Table 5 More productive authors

Number of documents	Author	Affiliation
32	Ozili P.K	University of Essex, Essex Business School, England
23	Bongomin, G.O.C	Makerere University, School of Business, Uganda
20	Ntayi, J. M	Makerere University, School of Business, Uganda
19	Asongu, S. A	University of South Africa, South Africa
18	Munene, J.C	Makerere University, School of Business, Uganda
16	Kassim, S	International Islamic University Malaysia, Malaysia
16	Mhlanga, D	University of Johannesburg, South Africa
14	Koomson, I	University of New England, UNE Business School, Australia
13	Makina, D	University of South Africa, South Africa
13	Ghosh, S	Reserve Bank of India, India
13	Sahu, T.N	Department of Commerce, India
13	Hassan, M.K	University of New Orleans, United States of America

Source: Authors' own work via Scopus data (2023)

in developing nations are observed. This finding is consistent with the overall scientific results.

The most cited authors in the field are shown in Table 6. To create this table, the most cited authors in Scopus and WoS were considered separately. Therefore, the table is a combination of the most cited authors in both databases. Asli Demirgürç-Kunt and Leora Klapper appear at the top of the list. These two researchers coauthored several studies, and together with Franklin Allen and Maria Soledad Martinez Peria, they also coauthored one of the most cited documents in the sample, Allen et al. (2016). Discrepancies exist between the list of the most productive and influential authors and that of the most productive countries, suggesting a weak relationship between productivity and impact. Additionally, a significant number of researchers affiliated with institutions in developed nations are observed. This contrasts with the most productive countries, which are led by China and India.

Table 6 Most cited authors

Number of citations	Number of documents	Author	Affiliations
987	6	Klapper, L	World Bank, United States of America
872	5	Demirgürç-Kunt, A	University of Pennsylvania, The Wharton School, United States of America
848	32	Ozili, P. K	University of Essex, Essex Business School, England
625	4	Allen, F	Imperial College of London, England
563	13	Hassan, M. K	University of New Orleans, United States of America
525	4	Blumenstock, J	University of California, Berkeley, United States of America
504	2	Peria, M.S.M	World Bank, United States of America
504	6	Weill, L	University of Strasbourg, Strasbourg Business School, France
499	5	Taghizadeh-Hesary, F	Tokai University, Japan
474	2	Sarma, M	Jawaharlal Nehru University, India
461	1	Pais, J	Institute for Studies in Industrial Development, India
409	14	Koomson, I	University of New England, UNE Business School, Australia
388	2	Le, T-H	Fulbright University, Vietnam
378	23	Bongomin, G. O. C	Makerere University, School of Business, Uganda

Source: Authors' own work via WoS and Scopus data (2023)

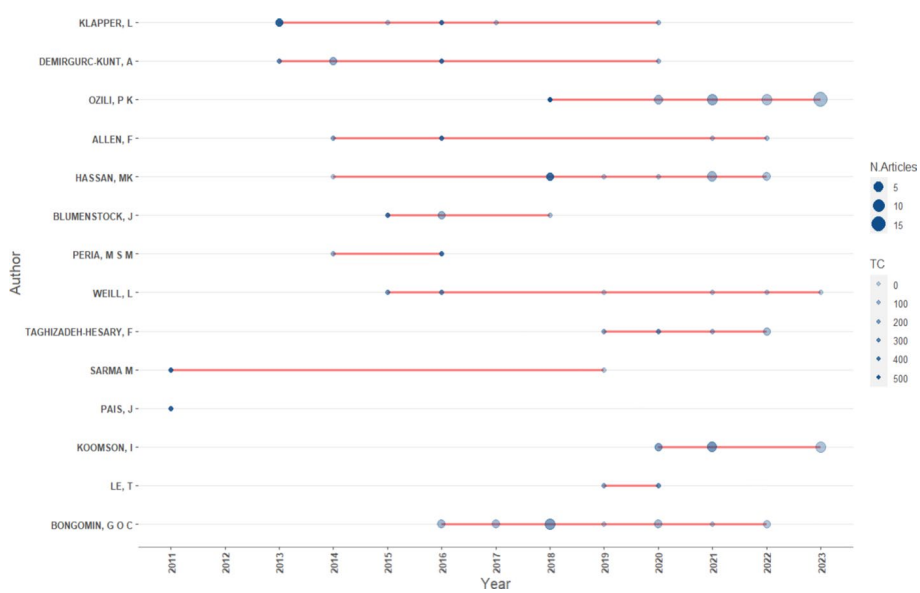

Fig. 4 Authors' production over time. *Source* Authors' own work via WoS and Scopus data (2023)

Figure 4 illustrates the evolution of the scientific production of the most cited authors, as outlined in Table 6. The size of the circles on the map corresponds to the number of documents, whereas the intensity of the color represents the number of citations. The analysis indicates that authors with higher citation counts consistently maintain substantial productivity over time. This suggests that they can be identified as experts in the field. The exceptions are Jesim Pais, Thai-Ha Le, and Maria Soledad Martinez Peria, who appear to be occasional contributors.

Intellectual structure

The references cited in the documents constituting the sample were analyzed to examine the intellectual structure used in the research field through cocitation analysis. The term "cocitation" is used to describe the frequency with which two documents are cited together in another document (Small 1973). This approach is often applied in bibliometric analyses to research the intellectual framework of the most significant documents in a certain field.

Figure 5 illustrates the cocitation network of the documents in the sample of this study. It was constructed using the references of all the documents, totaling 219,059 references. Most of the network consists of relatively recent documents from the period between 2008 and 2020. However, some older documents, such as those published by Stiglitz and Weiss (1981), McKinnon (1973), and Arellano and Bond (1991), are also included. None of the nodes stand out for their size, and the network can be described as homogeneous. In fact, there are only two clusters, and they are intertwined. The most extensive cluster, the red cluster, includes 40 documents that are related mainly to the measurement of FI (Sarma 2008; Demirgüç-Kunt and Klapper 2012; Demirgüç-Kunt and Kappler 2013; Chakravarty and Pal 2013; Demirgüç-Kunt et al. 2018), public policies (Claessens 2006; Allen et al. 2016; Ahamed and Mallick 2019), poverty (Chibba 2009; Omar and Inaba 2020; Beck et al. 2008; Dupas and Robinson 2013; Bruhn and Love 2014; Park and Mercado 2018), financial literacy (Lusardi and Mitchell 2014), banking services (Burgess and Pande 2005; Beck et al. 2007a, b), mobile money (Jack and Suri 2014; Suri and Jack 2016), and economic growth or development (Levine 2005; Sarma and Pais 2011; Sharma 2016; Kim et al. 2018; Mader 2018). The documents grouped in the green cluster include econometric documents (Pesaran 2007) and documents related to digitalization (Gabor and Brooks 2017) or sustainability (Le et al. 2019, 2020). As expected, some of the authors that appear in Table 6 can also be found in the network. This structure

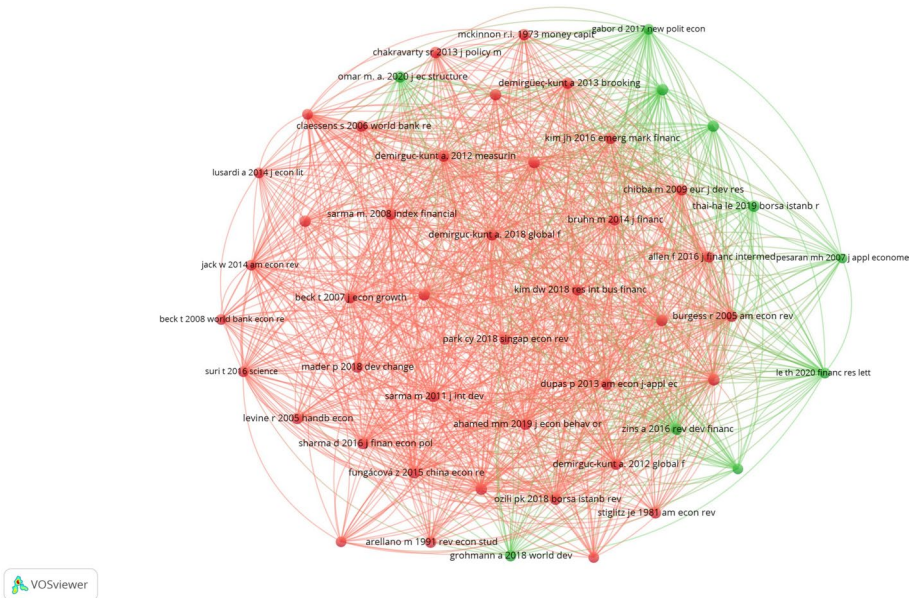

Fig. 5 Cocitation network of documents. *Source* Authors' own work via WoS and Scopus data (2023)

reflects the intellectual foundations of the field and aligns with the theoretical frameworks outlined in the section on theoretical foundations. It particularly concerns works on financial development and access under financial development theory and the emergence of tool-oriented literature in separate citation clusters that align with technology adoption models. The minimal cross-linkage between these clusters supports the observation that integration with the inclusive growth framework remains limited within the current bibliometric structure.

A word cloud representation was chosen to illustrate the evolution of the topics studied from 2011 to 2023. Figure 6 shows the word clouds of all the documents published in 2011, 2015, 2019, and 2023, which use words from abstracts. Standard stop words were removed from the sample. Words such as "financial" and "inclusion" were also eliminated because they could alter the results, considering that they were part of the original search.

Le et al. (2019) organized the literature on FI into three main parts: the construction of indicators for FI, the analysis of determinants influencing FI, and the exploration of the interconnections between FI and different dimensions of financial and economic development. Figure 6 illustrates the increase in documents published related to the latter line of research, as evidenced by the prominence of the words "economic" and "development" in the word clouds of 2019 and 2023 compared with earlier years. Interestingly, the word "countries" remains constant, which may indicate sustained interest over time in investigating determinants influencing FI, such as in Zins and Weill (2016) or Zeqiraj et al. (2022). Something similar happens with the word "services." Other noteworthy words include "banking" and "finance", which decrease over time, whereas "digital" and "growth" are attracting the attention of researchers. This result is in keeping with previous findings, which show research related to digital

Fig. 6 Word clouds of the abstracts of documents published in 2011, 2015, 2019, and 2023. *Source* Authors' own work using WoS and Scopus data (2023)

finance (Gabor and Brooks 2017; Ozili 2018) and growth (Kim et al. 2018; Usman et al. 2021) among the most cited documents. This progression reflects a shift from broad inclusion goals toward more tool-specific and institutional approaches, in line with the inclusive growth and technology adoption frameworks discussed earlier.

In Fig. 7, a multiple correspondence analysis (MCA) of the keywords illustrates the relationships between keywords by using distance on a two-dimensional plane. Keywords approaching the center have attracted increased attention in recent years (Xie et al. 2020).

The interpretation of the findings is based on the relative positions and distributions of points in relation to the dimensions. The closer the representation of words on the map is, the more similar their distribution is (Aria and Cuccurullo 2017). Three different clusters are identified, reflecting distinct lines of research. Cluster 1, in red, consists of four keywords and comprises documents on "financial service", "financial inclusion", "economics", and "finance." Therefore, the cluster emphasizes financial access and service-related terms, which aligns with financial development theory. Cluster 2, the blue cluster, is the most significant. This figure shows almost all the words related to the FITs for the proposed FI. This second cluster aligns with technology adoption models. The last cluster, the green cluster, is formed by nine keywords and comprises documents related to "income", "trade", "CO_2 emissions", and "China," suggesting links to broader macroeconomic and sustainability themes. Examples of the latter are found in Shahbaz et al. (2022) and Zhao et al. (2022). Considering the diversity in the sources where documents published in categories such as business, finance, economics, and environmental science are found, this cluster also stresses the different approaches adopted in research on FI. The MCA-based conceptual structure map shows that most FIT-related keywords are grouped within a single cluster, especially those related to mobile money, microfinance, and financial education. This suggests emerging convergence around these tools, which often cooccur in research focused on digital delivery, user capability, and community-based access models.

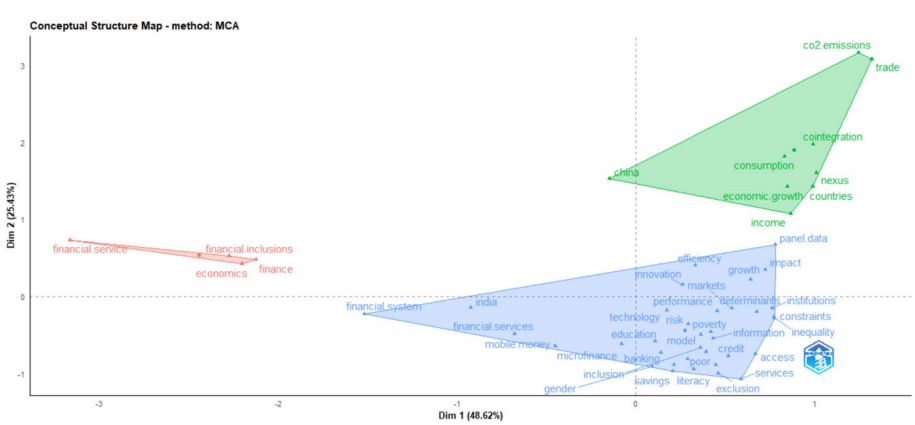

Fig. 7 Conceptual structure map using the MCA method. *Source* Authors' own work via WoS and Scopus data (2023)

Table 7 Financial inclusion innovation tools categories and keyword search

Financial innovation tools (FITs)							
FIT 1	**FIT 2**	**FIT 3**	**FIT 4**	**FIT 5**	**FIT 6**	**FIT 7**	**FIT 8**
Mobile money	Nonbank financial distribution	Behavioral credit risk	Workers' remittances	Adapted financial regulation	Financial education	Micro-finance business models	Blockchain technol-ogy
Keyword search in abstracts							
Mobile money	Network	Credit	Remittance	Regulation	Education	Microfi-nance	Blockchain
Prepaid	Agent	Microcredit		Law	Literacy	Microfi-nance	Crypto
E-money Digital finance		Micro-credit		Policies	Financial education	Mobile Banking	Bitcoin
		Micro credit		Policy	Financial literacy	Greenfield	
					Consumer trust	Upscaling	
Number of documents							
413	344	783	151	1,715	871	485	141

Source: Authors' own work via WoS and Scopus data (2023)

Table 8 Cooccurrences of FITs in abstracts

	FIT 2	**FIT 3**	**FIT 4**	**FIT 5**	**FIT 6**	**FIT 7**	**FIT 8**
FIT 1	78	51	34	154	62	28	11
FIT 2		54	23	128	78	50	27
FIT 3			26	315	145	156	17
FIT 4				61	35	11	17
FIT 5					376	166	40
FIT 6						77	11
FIT 7							3

Source: Authors' own work via WoS and Scopus data (2023)

Trends in FITs for FI

To understand the research trends in the eight FITs analyzed in this paper, documents were categorized in accordance with the different tools proposed. The process of identifying documents associated with each FIT involved conducting searches for keywords that characterize each FIT within the abstracts. The compilation of keywords is primarily based on Priyan et al. (2023) and García-Corral et al. (2022) and is presented in Table 7. Among the total sample, 3,153 documents, representing 75.04%, included at least one FIT, underscoring the relevant role of FITs in the FI research landscape.

Table 7 also shows the total number of documents per tool. Since documents can be classified into more than one group, the total number of documents does not necessarily correspond to the sample size. The documents that have the highest citation counts for each FIT are the following: Ozili (2018) stands out for FIT 1, Goyal and Kumar (2021) for FIT 2, Soederberg (2014) for FIT 3, Munyegera and Matsumoto (2016) for FIT 4,

Allen et al. (2016) for FIT 5, Sarma and Pais (2011) for FIT 6, Zins and Weill (2016) for FIT 7, and Chen and Bellavitis (2020) for FIT 8. Except for Soederberg (2014), all these documents appear in the cocitation analysis network.

Table 8 shows the number of cooccurrences of the FITs in the abstracts. More specifically, the number of abstracts containing two or more tools is shown. The results were normalized according to a chi-square criterion, allowing for a comparison between the different categories. The numbers that appear in red indicate abnormally high cooccurrences.

The values in Table 8 indicate that 78 documents show cooccurrences between FIT 1 and FIT 2, exceeding the expected count. The documents that explore the usage and distribution of mobile money services also consider nonbank financial distribution topics. This co-occurrence is found in the top 20 most cited documents per tool in Maurer (2012), Munyegera and Matsumoto (2016), Bongomin et al. (2018) and, Gbongli et al. (2019). Additionally, more cooccurrences than expected are found between FIT 5 and FIT 3, as seen in Chakravarty and Pal (2013) and Soederberg (2014). It seems that analyzing the adoption of innovative tools, such as behavioral scoring systems or decentralized authorization processes, is highly related to credit business models such as microcredit institutions. These new business models operate without proper regulation, prompting studies to focus on effective regulatory frameworks for these initiatives. This may also explain the cooccurrences between regulation and financial education, FIT 5 and FIT 6, as observed in Zins and Weill (2016), Park and Mercado (2018) and Goyal and Kumar (2021). Innovative business models, FIT 7, show higher-than-expected cooccurrences with alternative credit systems, FIT 3, including Guérin et al. (2013) and Mushtaq and Bruneau (2019). The adoption of behavioral scoring systems initially prevailed in microcredit and microfinance institutions.

The high frequency of mobile money and microfinance reflects their foundational role in the inclusion literature. However, the limited co-occurrence among different FITs indicates that research on these tools continues to be developed as separate lines of inquiry, with few studies examining their combined or comparative effects. This may reflect disciplinary silos, where different FITs are studied within distinct academic domains, limiting the development of integrated analytical frameworks. This fragmentation limits the potential to understand how innovation tools function simultaneously across economic systems as framed by financial development theory, user adoption environments as explained by technology adoption models, and equity-driven policy contexts emphasized in the inclusive growth framework. For instance, the weak linkage between mobile money (FIT 1) and regulatory adaptation (FIT 5) may point to a mismatch between rapid technological adoption and slower institutional adjustment, an imbalance highlighted in TAD models. Similarly, the limited overlap between financial education (FIT 6) and other tools may reflect gaps in inclusive growth strategies, where enabling policies fail to accompany innovation diffusion.

To facilitate the identification of the research trends in FITs, Fig. 8 presents the number of documents published. The eight FITs are represented as individual lines. FIT 5 shows the most pronounced exponential growth, followed by FIT 6 at a considerable distance. For almost all tools, an increasing trend is observed in the number of documents published up to the final year of the period considered. FIT 3 and FIT 4, which show a

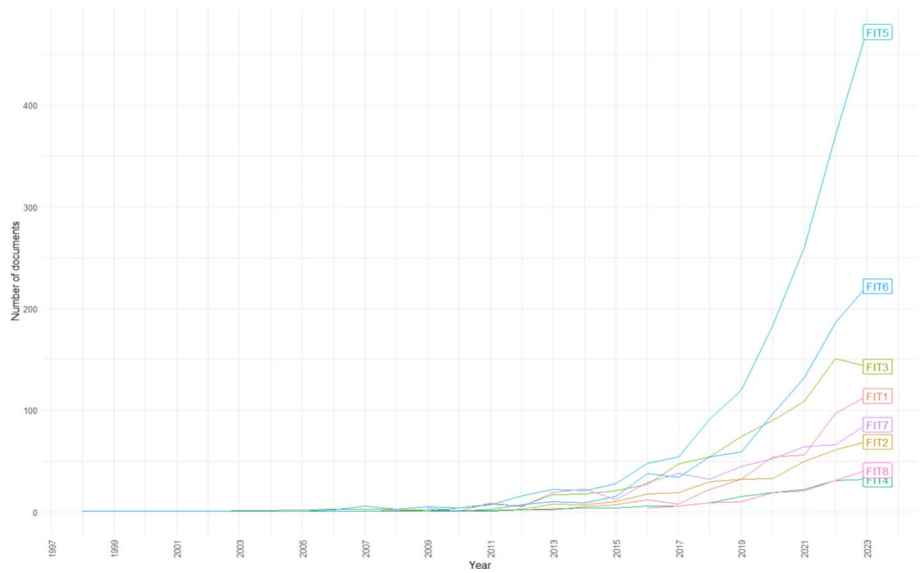

Fig. 8 Number of documents per FIT. *Source* Authors' own work via WoS and Scopus data (2023)

Table 9 FIT ranked

FIT	Number of documents	Average of citations per document	Average citation per document and year	Slope	Position
FIT 1: Mobile money	413	14.3	4.36	10.75	2
FIT 2: Nonbank financial distribution	344	14.1	3.10	6.18	4
FIT 3: Behavioral credit risk	783	11.0	2.60	14.35	5
FIT 4: Workers' remittances	151	9.9	2.38	3.43	8
FIT 5: Adapted financial regulation	1,715	11.8	3.39	42.53	1
FIT 6: Financial education	871	11.1	2.71	20.55	3
FIT 7: Microfinance business models	485	9.87	1.94	6.51	7
FIT 8: Blockchain technology	141	12.4	3.26	5.11	6

Source: Authors' own work via WoS and Scopus data (2023)

declining trend in 2023, are exceptions. Although the dataset does not cover all of 2023, its impact on the trends is believed to be minimal. The analysis suggests a concentration of researchers' efforts on specific FITs. Particularly noteworthy is the heightened focus on financial regulation, financial education, and mobile money, which are considered current hot research topics.

Table 9 complements the information in Fig. 8 by ranking the FITs studied. This ranking considers not only the total number of documents published per FIT but also the average number of citations per document, the average number of citations per document and year, and the growth in terms of publications measured by the slope of a regression model between the years and number of publications. When all these variables are considered, FIT 5 maintains its top position, and FIT 1 now ascends to the second position, a notable change from its fourth position in the preceding graph. Moreover, FIT6 maintains its significance. Their positions reflect both sustained scholarly

interest and increasing engagement with institutional mechanisms and digital or behavioral enablers of FI. In contrast, nonbank distribution and behavioral credit risk occupy mid-ranking positions, whereas blockchain, microfinance business models, and workers' remittances rank lower across most metrics. These differences highlight variation in research maturity and visibility across tools, reinforcing the value of a comparative, multi-indicator framework to assess their evolving role in the field. Despite their relevance, tools such as blockchain, remittances, and microfinance consistently rank lower across publication volume, citation averages, and growth metrics. Their marginal presence suggests underexplored opportunities to examine their evolving roles in FI.

Figure 9 shows the co-occurrence network of bigrams in the abstracts of documents related to the fastest-growing FITs depicted in Fig. 8: FIT 5 and FIT 6. The size of the node corresponds to the frequency of the bigram, and the size of the line that connects two nodes corresponds to the number of times these two bigrams occur together in an abstract. Additionally, the figure shows several clusters, each in a different color, grouping bigrams that tend to cooccur more frequently.

With regard to the bigram presented in Fig. 9-FIT 5, in which FI appears at the center of the network, the documents associated with this FIT predominantly establish connections between FI and "financial services." This term constitutes a fundamental aspect of its formal definition. Extending the analysis to other nodes, the documents that explore FI and regulatory aspects are also linked to "economic growth", "financial system", "financial literacy", and "financial institutions." Notably, the UN Sustainable Development Goals (SDGs) appear in these studies, as seen in the purple nodes "sustainable development" and "development goals." The latter are also shared with FIT 6, as shown in the bigram of Fig. 9-FIT 6. Terms such as "financial services", which relate to "digital finance," "sustainable development," "positive impact," and "economic growth" are also found here. Figure 9 also shows the connection of FI with a set of concepts related to the construction of indicators for FI, such as the "global index."

While FITs have expanded access to financial services, their impacts are not uniformly positive. In some contexts, mobile money and digital credit tools have been linked to overindebtedness, weak regulation, or exclusion of marginalized populations (Osabutey

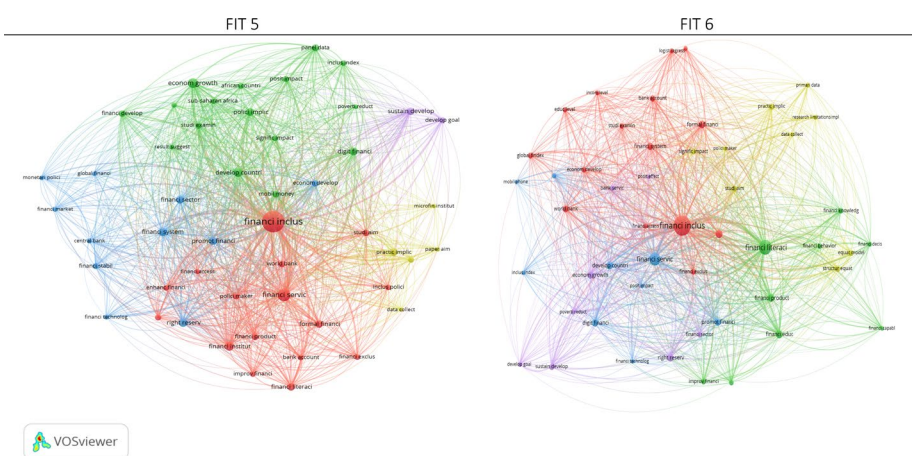

Fig. 9 Co-occurrence of words in abstracts. *Source* Authors' own work via WoS and Scopus data (2023)

and Jackson 2024). Moreover, users may adopt digital financial tools even when they do not fully understand associated risks, particularly in environments with low digital literacy (Senyo and Osabutey 2020). These challenges underscore the importance of context-aware policy frameworks and responsible innovation to ensure that FITs promote inclusion without reinforcing inequalities.

This study provides the most comprehensive bibliometric mapping of FITs to date, integrating major databases and linking findings to theoretical and policy frameworks. The results offer a clear overview of how research on FI and FITs has evolved. Performance analysis, cocitation, keyword mapping, and tool-specific rankings identify both active and emerging research areas. The findings show that FITs are often studied separately, revealing a fragmented field. This fragmentation limits the understanding of how innovation tools interact across financial systems and reflects weak integration between financial development, technology adoption, and inclusive growth frameworks. The findings reinforce the relevance of these frameworks by explaining how FITs operate across institutional and socioeconomic contexts. This underscores the need for integrated research connecting institutional, technological, and policy perspectives on FI to support more effective, scalable, and context-sensitive policies and innovation strategies.

Discussion and conclusions

This study uses bibliometric and visualization techniques to evaluate global trends in FI and innovative tools facilitating its dissemination. A comprehensive merged database consisting of 4,202 documents, including research articles, book chapters, and conference papers, was created. The dataset was compiled from Scopus and WoS, the two most widely used databases for bibliometric analysis. Merging these databases allowed for a more powerful analysis and included relevant documents unique to either Scopus or WoS. This approach ensures a more comprehensive and accurate dataset for bibliometric studies. Most of the highly cited documents appear in either WoS or Scopus. Additionally, the use of book chapters has proven to be a relevant source for disseminating research in this field. Notably, one of the most cited documents during the period analyzed is a book chapter. It is therefore advisable to consider merging databases and incorporating book chapters when reviewing the FI literature.

Expanding on bibliometric analysis, this study also explores trends in tools promoting FI, a rapidly growing research field. Key topics include financial regulation, financial education, and mobile money, whereas interest in behavioral credit risk, although still the third most researched topic, is declining. Researchers should note that some topics frequently cooccur, such as mobile money with nonbank distribution, financial regulation with behavioral credit risk, and regulation with financial education. Additionally, behavioral credit risk often overlaps with microfinance business models. Researchers should consider these concurrences.

Research on FI has been conducted from a multidisciplinary perspective. This results in the dissemination of research findings across journals that may adopt very different approaches. Given the exponential growth in scientific production on this topic, it would be beneficial for publishers to establish a journal dedicated to consolidating the publication of results. This would also be useful for researchers, as they would have a primary source of documentation to refer to.

This bibliometric analysis enhances the understanding of the research landscape, informs policy decisions and industry practices, and guides future research in the fields of financial innovation and FI. The following research questions, the answers of which are presented in this concluding section, were developed in this study.

RQ1: What are the current publication trends, the most influential documents, sources, and authors in FI?

The research trajectory on FI shows an exponential growth in the scientific production over time, highlighting its significance. The annual growth rate during the period studied (1998–2023) was 28.59%. This percentage significantly exceeds the average growth rate of scientific publications, estimated at 4%. The body of knowledge expanded rapidly between 2010 and 2012 and has continued to grow ever since, with China and India leading in research production.

The most influential document, which analyses the impact of digital finance on FI, is the study carried out by Ozili (2018). Another significant study in the field is that of Gabor and Brooks (2017). However, in terms of citations per year, the most cited document is a literature review on financial literacy by Goyal and Kumar (2021), followed by a study of the impact of FI on economic growth (Usman et al. 2021).

Peterson K. Ozili is the most productive author in a field where most authors have published only one study. Most of the documents are written in collaboration with at least one other researcher, and almost one third of the documents involve international collaboration. Leora Klapper and Asli Demirgürç-Kunt are at the top of the list of the most influential authors. Despite the fact that developing countries produce more documents, developed nations are currently leading the discourse on the topic. Future research could examine how the substantial scientific production generated by India, China, and other developing countries may influence or reshape this dynamic.

The journal most frequently employed by authors to disseminate their findings is *Sustainability*. It has emerged as the leading source, with 103 articles published on the topic, reflecting its significant role in academic discussions. Because of its multidisciplinary nature, it is selected as a publishing outlet for researchers from different disciplines that analyze FI. Nevertheless, it is worth noting that none of the most cited documents are published in this source. When more specific works related to economics are explored, other notable journals worth considering are the *International Journal of Social Economics* and *World Development*.

The results of this study are difficult to compare with those of other bibliometric studies on FI. This is because, on the one hand, a unique database was used, and on the other hand, not only research papers but also book chapters and conference papers were considered. However, the findings of this research regarding the most productive author, the most influential documents, and the sources differ slightly from those of previous research. Compared with prior Scopus-based studies (Chhatoi et al. 2021; Suryavanshi et al. 2024), the merged dataset confirms the prominence of authors such as Klapper and Demirgüç-Kunt. It also highlights influential and productive contributors such as Ozili, Blumenstock, and Bongomin, who receive less attention in single-source analyses. In terms of publication sources, while earlier studies emphasize journals such as the *International Journal of Social Economics*, our results identify *Sustainability*, *World*

Development, and the *Journal of International Development* as leading in both citations and volume. The top-cited documents in the dataset include works by Ozili (2018), Sarma and Pais (2011), and Allen et al. (2016), reflecting a shift toward innovation- and policy-focused research. The geographic analysis confirms the leading role of the United States, China, and India but also shows increased representation from countries such as the United Kingdom, Uganda, and South Africa. These differences illustrate how merging databases and expanding source types offers a broader, more representative view of the FI research landscape.

RQ2: What is the intellectual structure of the literature on FI?

This study examines the intellectual structure of the literature through cocitation analysis of references, word cloud analysis of abstracts, and multiple correspondence analysis of keywords. The multiple correspondence analysis identifies three main clusters: the first comprises documents focused on "financial services", "economics", "finance", and "financial inclusion"; the second includes documents on econometrics, the "financial system" and "financial services", encompassing FITs; and the third contains documents related to "CO_2 emissions", "China" and "income." Furthermore, cocitation analysis indicates that works centered on sustainability and digitalization are different from those addressing the measurement of FI, public policies, poverty, financial literacy, banking services, mobile money, and economic growth or development. The word cloud analysis stresses a growing body of literature exploring the interconnection between FI and various dimensions of financial and economic development. Together, these patterns reflect the influence of the three theoretical frameworks guiding this study: financial development theory, technology adoption models, and the inclusive growth framework.

The analysis of article abstracts over time reveals the growing prominence of "economic" and "development." While terms such as "banking", "housing", and "finance" are declining, "digital" and "development" are increasing, aligning with trends in digital finance, the SDGs, and growth research. Moreover, cocitation and keyword mapping reveal limited integration across thematic clusters, highlighting a fragmented intellectual structure that reflects the separation of FIT-related research streams.

RQ3: What are the research trends in FITs that promote FI?

This research differs from other studies, as an in-depth analysis of the scientific production and trends of FITs aimed at promoting FI is carried out. This paper ranks the FITs used to promote FI from three different perspectives: first, by analyzing the frequency of FIT cooccurrences in abstracts; second, by featuring the trend of each FIT; and third, by considering both the number of documents per FIT and the average number of citations per document. The structured categorization of eight FITs also offers a replicable framework for future bibliometric or empirical studies in the field.

While considering that more than one FIT is not uncommon in the FI literature, cooccurrence analysis reveals that certain FITs are often considered together. For example, documents discussing mobile money are likely to also consider nonbank financial distribution. Similarly, documents addressing regulation often cover financial education. Those analyzing behavioral credit risk typically also explore innovative business models and regulatory frameworks. The trend in the number of documents published also

illustrates that the development of regulatory frameworks is the FIT predominantly explored, followed by mobile money and financial education. These three tools represent areas where researchers are currently investing considerable efforts. The results of this study are consistent when considering both the number of documents per FIT and the average number of citations per document. In contrast, blockchain, remittances, and microfinance receive comparatively less attention across all indicators, suggesting that their evolving role in FI remains largely unexplored.

RQ4: What are the potential areas for future research?

Upon completing this bibliometric study, several potential areas for future research were identified. Future investigations should explore these areas to both complement and expand on the existing body of research.

First, the results obtained by combining the responses to RQ2 and RQ3 indicate that research in recent years has shifted in direction. Within this context, future studies should explore the connection between the three most prominent FITs and digitalization, economic growth, development, and sustainability.

Second, although developing countries currently lead scientific production, developed nations remain at the forefront of the discourse on the topic. It would be worth analyzing how the significant scientific output from countries such as India and China might influence or reshape this dynamic in the future.

Third, researchers should evaluate whether new FITs should be included in the analysis. It would also be interesting to study the causal impact of these tools on promoting FI to understand their effects and assess their effectiveness for this purpose. The incorporation of these elements could significantly increase the depth and relevance of future studies in the field.

Finally, relatively few studies have examined the potential of remittances to increase FI. This area should be further explored, as remittances may play a significant role in improving access to financial services, particularly in underserved regions. Investigating how remittances contribute to FI could yield insights into how these financial flows can be leveraged more effectively to promote broader economic participation and stability. Addressing this research gap could have important implications for policy-making and financial service development. Future studies should also adopt integrative approaches that analyze how multiple FITs operate jointly within institutional, technological, and policy environments.

Implications

This study is important for various stakeholders in the FI landscape. It provides a systematic and comprehensive review of the current literature, offering critical insights that benefit academics, policymakers, and industry practitioners.

Academics may find this analysis valuable, as it identifies potential research gaps, underexplored areas, and research ideas needing further study. Additionally, it places emphasis on leading authors, journals, and seminal works that could inspire new research ideas and collaborations. Given the observed fragmentation in FIT-related research, future academic work should also aim to develop integrated frameworks that examine how different tools interact in diverse institutional settings. Moreover, the

increasing research output from developing economies offers an opportunity to rebalance the global FI discourse. Collaboration across regions could ensure that policy and innovation reflect broader realities.

For policymakers, this bibliometric analysis provides an evidence-based understanding of the current research landscape. This allows them to assimilate key findings and trends, highlighting the critical role that FITs play in enhancing financial accessibility and economic development. This thorough understanding can guide the development of targeted policies and interventions aimed at promoting FI. Furthermore, given that "Adapted Financial Regulation" has emerged as the most relevant FIT, policymakers should be recognized as having a prominent role and responsibility in encouraging FI initiatives. Additionally, "Financial Education" also appears in this study to be a particularly important area that policymakers should address to promote FI properly. These findings can support the development of aligned regulatory strategies that not only promote FI but also contribute to broader SDGs, particularly in underserved regions. Tools such as remittances, blockchain, and microfinance remain underutilized in policy design, despite their potential for advancing inclusion in migrant, rural, or low-infrastructure contexts.

Industry practitioners, including companies, traditional financial institutions, and nongovernmental organizations, can gain insights into the current state and potential impact of FITs on FI. Moreover, being aware of how FITs might improve FI should be the driving force behind the development of more innovative solutions. Among the different tools studied, "Mobile Money", "Nonbank Financial Distribution" networks, and "Behavioral Credit Risk" systems are the FITs researchers found to be the most relevant to be used by practitioners to improve FI.

Limitations

This research on the use of FITs to improve FI has several limitations that should be acknowledged.

The main limitation of this paper arises from an issue that is common in bibliometric analysis: the identification of authors. The Scopus Author ID was chosen to compile information on the most productive and most cited authors. However, since this identifier does not exist in the WoS database, 461 documents were excluded. To ensure that this omission did not impact the overall representation of the most productive and most cited authors, a manual check was conducted on the impact information of the omitted authors. While this verification mitigates this issue, it might not entirely eliminate the potential bias created by excluding these documents.

Classifying papers heavily depends on the keywords used. Different keywords could lead to different results. This limitation suggests that the findings of the study would have varied if alternative or additional keywords had been employed. This could affect the comprehensiveness and accuracy of bibliometric analysis. Additionally, the study focuses on eight predefined FITs, which, while widely cited, do not capture the full range of emerging financial technologies or region-specific innovations.

Owing to the length constraints of the paper, most papers could not be discussed in depth. This limitation means that while this study provides a broad overview of the

literature, it might lack detailed insights into specific papers or discussions of their contributions.

Finally, the use of different software instruments or even different algorithms to generate network maps could provide slightly different results. This limitation highlights the potential variability in the visual representation and interpretation of bibliometric data on the basis of the tools used. Inferences drawn from clustering techniques such as co-occurrence or MCA should also be interpreted cautiously, as the results are sensitive to parameter selection, proximity thresholds, and keyword normalization.

Abbreviations

DOI	Digital object identifier
ESCI	Emerging sources citation index
FI	Financial inclusion
FIT	Financial innovation tool
JCI	Journal citation index
JCR	Journal citation reports
JIF	Journal impact factor
MS Excel	Microsoft excel
MCA	Multiple correspondence analysis
Q	Quartile
RQ	Research question
SCIE	Science citation index expanded
SJR	Scimago journal and country rank
SSCI	Social science citation index
SDG	Sustainable development goals
TAM	Technology acceptance model
UTAUT	Unified theory of acceptance and use of technology
WoS	Web of science

Acknowledgements
We would like to express our sincere gratitude to the reviewers for their invaluable contributions to this study. Their expert feedback and constructive comments have greatly enhanced the quality of our research. We are very thankful for their support.

Author contributions
The authors contributed equally to the paper.

Funding
This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Availability of data and materials
The data that support the findings of the study are available on request from the corresponding author.

Declarations

Competing interests
The authors have no competing interests to declare.

Received: 2 January 2024 Accepted: 6 November 2025
Published online: 19 November 2025

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